



VISER: A Tractable Solution Concept for Games with Information Asymmetry

Jeremy McMahan, Young Wu, Yudong Chen, Xiaojin Zhu, Qiaomin Xie
University of Wisconsin-Madison

Introduction

Two players play a bimatrix game (A, B) with information asymmetry: the first player (*victim*) only knows A but the second (*exploiter*) knows both A and B .

Q: How will the game play out?

Captures security applications like adversarial RL.

Assumption 1: Both players seek to optimize their worst-case payoff with respect to the information available.

Assumption 1 is justified by the observation that real-world players are *uncertainty averse*: they do not take risks when missing full information.

Rational Behavior

The **victim**, which has an information disadvantage, cannot anticipate what the exploiter might do, so should play a *security strategy*:

$$x^* \in X^* := \arg \max_{x \in \Delta(n)} \min_{y \in \Delta(m)} x^T A y$$

The **exploiter**, which has an information advantage, wishes to leverage its advantage to achieve higher payoff than playing a security strategy.

- The exploiter can reason that the victim plays some security strategy $x^* \in X^*$.
- However, the exploiter does not know which x^* the victim selects, so suffers *strategic uncertainty*.

The exploiter should assume the worst-possible x^* is chosen, and so play a *worst-case best-response*:

$$y^* \in BR(X^*) := \arg \max_{y \in \Delta(m)} \min_{x^* \in X^*} x^{*T} B y$$

VISER

- Given the information asymmetry, standard solution concepts like NE and ROE no longer are defined.
- Hence, we propose a new solution concept called VISER (victim is secure and exploiter best-responds).

Definition 1: $VISER := X^* \times BR(X^*)$.

Theorem 1: VISER satisfies the following properties:

- A VISER solution always exists.
- VISER solutions are independent strategies, so can be played in a distributed fashion.
- Players who deviate from a VISER solution receive lower worst-case payoff.

Thus, **under Assumption 1, a VISER solution is played.**

Algorithms

The **victim** can compute its VISER strategy as a solution to the classic LP for computing NE in zero-sum games.

$$\begin{aligned} \max_{x \in \mathbb{R}^n, z \in \mathbb{R}} \quad & z \\ \text{s.t.} \quad & z \leq x^T A e_j, \quad \forall j \in [m] \\ & 1^T x = 1, \quad x \geq 0. \end{aligned}$$

The **exploiter** can first simulate the victim's LP and use the objective value z^* to compute its VISER strategy.

$$\begin{aligned} \max_{x \in \mathbb{R}^n, z \in \mathbb{R}} \quad & z^* 1^T w - \alpha \\ \text{s.t.} \quad & \alpha + e_i^T B y - e_i^T A w \geq 0 \quad \forall i \in [n] \\ & 1^T y = 1, \quad y \geq 0 \quad w \geq 0. \end{aligned}$$

The key to deriving this LP is applying LP duality on the exploiter's inner min to reveal the linear structure.

Theorem 2: Both players can compute a VISER strategy independently in polynomial time.

Markov Games

To study solutions for Markov games, we define Markov-perfect VISER (MPVISER) by,

- **Victim:** $\Pi_{hs}^* := \arg \max_{\pi \in \Pi} \min_{\nu \in N} V_{hv}^{\pi, \nu}(s)$
- **Exploiter:** $N_{hs}^* := \arg \max_{\nu \in N} \min_{\pi \in \Pi_{hs}^* \cap \bigcap_{k>h,s'} \Pi_{ks'}^*} V_{he}^{\pi, \nu}(s)$

Definition 2: $MPVISER := \bigcap \Pi_{hs}^* \times \bigcap N_{hs}^*$

Theorem 3: MPVISER can be computed in poly time.

Experiments

